

End to End Testing

End to end testing (E2E testing) is a software testing method that involves testing an application's workflow from beginning to end. It aims to replicate real user scenarios to validate the system for integration and data integrity. The test goes through every operation the application can perform. It tests how the application communicates with hardware, network connectivity, databases, and other applications.

End to end testing determines if various dependencies of an application are working accurately. It also checks if accurate information is being communicated between multiple system components.

Importance of E2E Testing:

- **Backend:** Verifies the database and backend layers of an app. This is necessary since the app's core functionalities depend on backend capabilities.
- **Multi-tier system:** If an application has a complex architecture whose workflow works through multiple tiers, E2E testing is verifies overall functions and the interaction between individual tiers in the architecture.
- **Distributed Environment:** E2E testing is necessary if an application is based on service-oriented architecture or cloud environments.
- **Consistent User Experience:** Since E2E testing involves the frontend, it also ensures that the app provides a user experience that works across multiple devices, platforms, and environments. Cross browser compatibility testing.

